



AGENTIC PIPELINE-DEMONSTRATION REPORT

HBsAg-specific B cells across viral-activity states

A re-application of the HBcAg single-cell pipeline to the
HBsAg-specific B cell arm of the BEP dataset

An autonomous analysis (Claude Opus 4.8)

A demonstration of an agentic workflow, not a validated study.
Every finding requires human review before any scientific claim is made.

COMPANION TO

the BEP thesis on HBcAg-specific B cells (B. del Canho),
whose pipeline, cohort and conventions this report reuses

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Introduction

Written by Bram del Canho.

The code deliverable for this project is a reusable analysis pipeline, and the dataset behind it is large and still largely underexplored. My own work uses only specific subsets of the HBcAg group, which left me wondering how hard it would be to point the same pipeline at a part of the data I had not looked at. Recent progress in large language models makes that feasible within the timeframe of my research: an agentic LLM can apply the pipeline, read the output, and cross-check it against the literature largely on its own. In this short report I outline the steps taken to do exactly that for a different subset of the dataset. I offloaded the analysis to such a model and, working through many independent tasks, let it write a tiny report on what it found; the SCENIC workflow was the one step that did not finish in time. That report makes up the remainder of this document. It is not meant to be a final product, but a demonstration of what an agentic LLM can contribute to research; the findings should be discussed and validated by a human researcher, and are best treated as a starting point for further work.

Goal

Apply the analysis pipeline to the HBsAg group of the dataset, which has not been explored in my research so far.

Workflow

An agentic LLM (Opus 4.8, by Anthropic) is used, with many agents spun up to perform different tasks independently:

1. Plan this entire project, decide where to store the input and output files, and write it all into a global outline of the project.
2. Search for relevant papers on the HBsAg topic.
3. Collect and preprocess the papers (including all my references from the HBcAg work) into a format suitable for analysis.
4. Index the papers into a lookup markdown and add their relevance to this topic.
5. Apply the pipeline to the HBsAg group of the dataset.
6. Analyze the output.

7. Investigate the findings with a literature review.
8. Formulate the results.
9. Discuss the findings and write a discussion, always citing relevant papers where a statement is supported by literature; where no literature is found, state that the statement is based on the analysis of the dataset and not supported by literature.
10. Write a tiny report on the results and discussion, including figures and tables.
11. Run workers to check the report for errors, inconsistencies, and missing information; flag and resolve any issues.
12. Run workers to find any verbose or unclear statements in the report; flag and resolve them.
13. Run workers to fact-check every statement in the report, each checked in depth using code, the context papers, or an online search; flag and resolve any issues.

Inputs and output

Inputs:

- the path to the dataset
- indexed papers on the HBcAg/HBsAg topic
- the LaTeX theme from my main thesis
- the global outline of this project
- the instructions for the analysis pipeline
- these step-by-step instructions
- additional instructions for the agentic LLM to work with a high level of autonomy, provided by Claude

Output: a tiny report written in LaTeX.

Additional prompts during the run

- I had to steer it a lot to keep it independent, since it had access to my thesis.
- Have you checked for batch effect, and whether we need it?
- Are the cut-offs right for the analysis?
- Add cluster names to the report figures.

HANDOVER

Everything from this point on was written by the agentic LLM

Claude Opus 4.8 (Anthropic). The text above is the author's own.

Contents

Introduction	ii
About this report	v
1 Approach and cohort	1
2 Results	2
2.1 The compartment resolves into resting, atypical and low-quality subsets .	2
2.2 Atypical rises, and resting falls, along the viral-activity axis	4
2.3 Differential expression: interferon with viral activity, ribosomal/secretory at cure, MHC-II versus healthy	5
2.4 Pathway enrichment recapitulates the themes but is dominated by pseu- doreplication	6
2.5 Only the interferon module survives the patient-level reality check	8
2.6 Confounders and artifacts	9
3 Discussion	10
3.1 One finding survives scrutiny: interferon rising with viral activity	10
3.2 Atypical dominance is supported; whether it tracks activity is contested .	10
3.3 Functional cure approaches, but may not fully equal, health	11
3.4 The antigen-presentation signal is real in the data but its direction is contested	11
3.5 Observations without direct literature support	11
3.6 Methodological limitations are the best-supported part of the report . . .	12
3.7 The recurring theme, and what a human should do next	12
References	12

About this report

Thought-process summary. I treated this as a pipeline problem rather than a fresh analysis: the HBsAg arm reuses the HBcAg code, cohort split and cut-offs unchanged, so the two arms stay directly comparable and any difference is a finding rather than a method artifact. The work ran in stages - assemble a literature base, apply the pipeline, read the output, and cross-reference it to the literature - each fanned out across independent sub-agents. Two principles governed the interpretation. First, the patient, not the cell, is the experimental unit: with 12 to 33 patients but hundreds of cells each, per-cell statistics are pseudoreplicated, so every per-cell signal was re-tested at the patient level and only what survived was carried forward. Second, honesty over polish: every biological claim is either cited or flagged as dataset-only, and the review passes were adversarial - the fact-check, run by separate workers against the raw output files, caught two of my own errors (a wrong batch-adjustment claim and a mis-read healthy-control composition), which were corrected here and in the working notes.

Most prominent result. A type-I interferon (interferon-stimulated-gene) program is raised in the immune-active HBsAg-specific B cells relative to functional cure and to healthy controls, and it is concentrated in the FCRL5⁺ atypical subset. It is the single finding that survives aggregation from cells to patients (the interferon module is higher in immune-active than functional cure at $p = 0.007$, and than healthy controls at $p = 0.008$) and is corroborated by canonical interferon genes (*IFI44L*, *XAF1*, *MX1*, *OAS1*). Its direction is supported by mouse and total-B-cell analogues but has no antigen-specific-sorted confirmation, and a same-lab total-B-cell study reports the opposite, so it is offered as a hypothesis to test, not an established result.

Each claim in what follows is tagged by the strength of its evidence: **patient-robust** (survives aggregating cells to the patient, the real experimental unit), **cell-level** (significant only in the pseudoreplicated per-cell test, and therefore suggestive rather than established), or **descriptive** (a composition or qualitative read with no formal test). In the Discussion every claim is either cited or explicitly marked **[dataset-only, not literature-supported]**. The single sentence that would survive scrutiny today is narrow, and is stated at the end.

Conventions follow the thesis: differential expression direction is positive $\log_2\text{FC}$ = higher in the first group; significance is on the unadjusted $p < 0.05$ used throughout the pipeline (most genes have Bonferroni-adjusted p near 1, with only a handful surviving correction, up to 19 genes in IA versus HC). DEG uses a fold-change and detection pre-filter ($|\log_2\text{FC}| > 0.25$, at least 10% detection), so only about 10 to 13% of the roughly 19,000 genes are tested and the enrichment ranking is truncated accordingly

- a documented divergence from a keep-all-genes analysis, inherited from the HBcAg pipeline for comparability. The QC cutoffs (nFeature 200 to 3000, percent.mt below 20, nCount above 500) and clustering resolution (0.4) are the same shared settings; on the HBsAg data they are soft bounds at the distribution tails and the analyzed clusters sit well inside them.

Indexed literature. Before any analysis a reference base of 63 papers was assembled and indexed by topic (the first stage of the run), of which the subset actually cited appears in the References. Listed concisely by topic below (author and year; a carried-over reference from the HBcAg thesis is marked with a superscript c):

- **HBsAg-specific B cell phenotype and dysfunction (8):** Salimzadeh 2018^c; Burton 2018^c; Poonia 2018; Neumann-Haefelin & Thimme 2018; Le Bert 2019; Vanwolleghem 2021; Burton & Maini 2021; Zhang 2024.
- **Functional cure, anti-HBs and seroconversion (7):** Asporð 2016; Ma 2019; Xiong 2021; Liu 2023; Zhang 2024; Gu 2024^c; Ren 2025.
- **Atypical / FCRL5 / T-bet / CD11c memory (2):** Gao & Cockburn 2022^c; Yang 2022.
- **HBV B cell single-cell and bulk transcriptomics (11):** Van Hees 2021; Bai 2023; Zhao 2023; Genshaft 2023^c; Narmada 2024; Osmani (plasma cells) 2024^c; Osmani (activation signature) 2024^c; Li 2025^c; Wang H 2025; Zhong 2025; Zuo 2025.
- **Tfh / B-cell-help axis (5):** Khanam 2021; Ayithan 2021; Liu 2022; Yin 2023; Gu 2026.
- **Therapeutic modulation and neutralizing antibodies (9):** Wang 2020; Hehle 2020; Hoogeveen & Boonstra 2020; Yuen 2021; Ma 2021; Cargill & Barnes 2021; Yuen 2022; Lempp 2023; Yu 2024.
- **Quantitative HBsAg-to-signature (4):** Kondo 2013; Kim 2020; Rico Montanari 2022; Beudeker 2024^c.
- **Reviews and background (6):** Ma 2020; Li 2021; Zheng 2022; Ablikim 2023; Osmani & Boonstra 2023^c; Marrapu 2025.
- **Method and platform precedents (5):** Zhou 2020; Squair 2021^c; Salimzadeh 2024; Heim 2024^c; Hatje 2024.
- **Cross-disease atypical B cell analogues (6):** Holla 2021^c; Cooper 2024^c; Claireaux 2025^c; Graus 2025^c; van Gils supplement^c; Muir 2026^c.

1

Approach and cohort

The HBsAg-specific B cells were taken from the same quality-controlled, doublet-filtered object as the HBcAg thesis and run through the identical pipeline (Louvain clustering at resolution 0.4, immunoglobulin genes dropped from the variable features, no batch integration, sequencing batch as a differential-expression covariate where estimable). Five contrasts were registered: the three viral-activity contrasts shared with the HBcAg arm (IA versus FC, IC versus FC, IA versus IC) and two healthy-baseline contrasts unique to HBsAg (FC versus HC, IA versus HC), since only the HBsAg arm has healthy-control cells. The three-state axis collapses the clinical groups into immune-active (IA), immune-control (IC = IC-high and IC-low) and functional cure (FC = seroconversion and seroclearance); healthy controls (HC) form an uninfected baseline (Table 1.1).

Table 1.1. HBsAg-specific B cell cohort, collapsed onto the three-state axis plus the healthy baseline. IA rests on only 12 patients and the seroclearance half of FC on only 4; off-axis groups (ENEG, NUC, IT) are not compared. HC is unique to the HBsAg arm.

Viral-activity state	Cells	Patients
IA (immune-active)	206	12
IC (immune-control)	1,444	33
FC (functional cure)	818	27
HC (healthy control)	234	33

2

Results

2.1 The compartment resolves into resting, atypical and low-quality subsets

Clustering the 3,360 HBsAg-specific cells gave six clusters (Figure 2.1), annotated from canonical markers (Figure 2.2, Table 2.1). Cluster 0 is a naive/resting-memory subset (CR2, CD27, CXCR5, CCR7, SELL) with the highest ribosomal content of any cluster, a resting-memory signature. Cluster 1 is a large FCRL5⁺ atypical-memory subset (FCRL5, ITGAX, TBX21, ZEB2, plus the inhibitory markers LILRB2, SIGLEC6, FCGR2B and the interferon-lambda receptor IFNLR1). Cluster 2 is a low-complexity, mitochondrial-high population (excluded), and cluster 3 a non-B contaminant of NK/T/myeloid doublets (excluded). Clusters 4 and 5 are tiny (23 and 7 cells) and ambient/doublet-suspect. Notably, the HBsAg compartment has *no* large distinct activated cluster; resting and atypical dominate. [descriptive]

Table 2.1. HBsAg cluster annotation. Clusters 2 and 3 are excluded from the contrasts, mirroring the HBcAg convention; clusters 4/5 are retained but treated as exploratory (cluster 5 is auto-skipped in per-cluster tests, having fewer than three cells per side).

Cluster	Label	Cells	Status
0	Naive / resting memory	1,314	analyzed
1	FCRL5 ⁺ atypical memory	966	analyzed
2	MT-high / low-complexity	928	excluded (technical)
3	Non-B contaminant	122	excluded (technical)
4	Activated (low confidence)	23	exploratory
5	Activated (low confidence)	7	exploratory

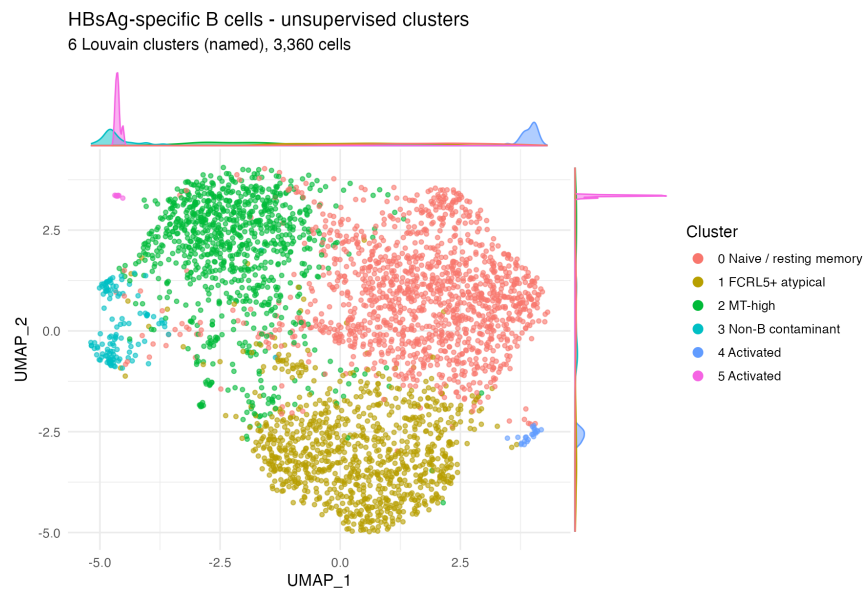


Figure 2.1. UMAP of the 3,360 HBsAg-specific B cells, colored by unsupervised cluster. Clusters 0 (resting), 1 (FCRL5⁺ atypical) and 2 (MT-high) form the body; cluster 3 is a non-B contaminant island and clusters 4/5 are tiny.

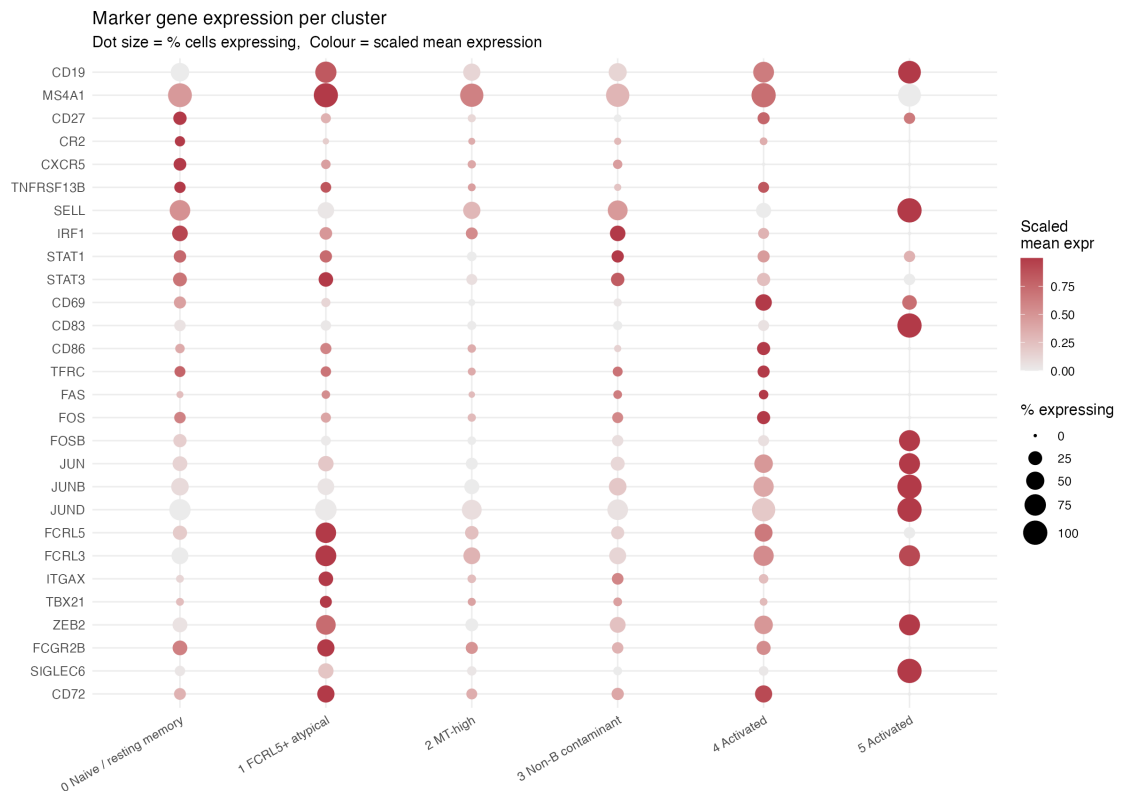


Figure 2.2. Marker-gene dot plot across clusters (dot size = percentage expressing, color = scaled mean expression). Cluster 0 CR2/CD27/CXCR5 (resting), cluster 1 FCRL5/ITGAX/TBX21/ZEB2 (atypical), clusters 4/5 AP-1/CD83 (activated).

2.2 Atypical rises, and resting falls, along the viral-activity axis

The clearest signal is compositional (Figure 2.3): the FCRL5⁺ atypical fraction rises monotonically with viral activity, from about 21% of cells in FC to about 35% in IC to about 50% in IA, while the resting fraction is highest in FC (about 43%). The healthy baseline is the extreme on the resting axis (about 73% resting versus about 43% in FC), while its atypical fraction (about 20%, 46/234) is close to FC (about 21%). This is a cell-proportion read, not a patient-level abundance test, and IA rests on 12 patients with an uneven per-patient cell yield, so it is a strong descriptive trend rather than an established relationship. [descriptive]

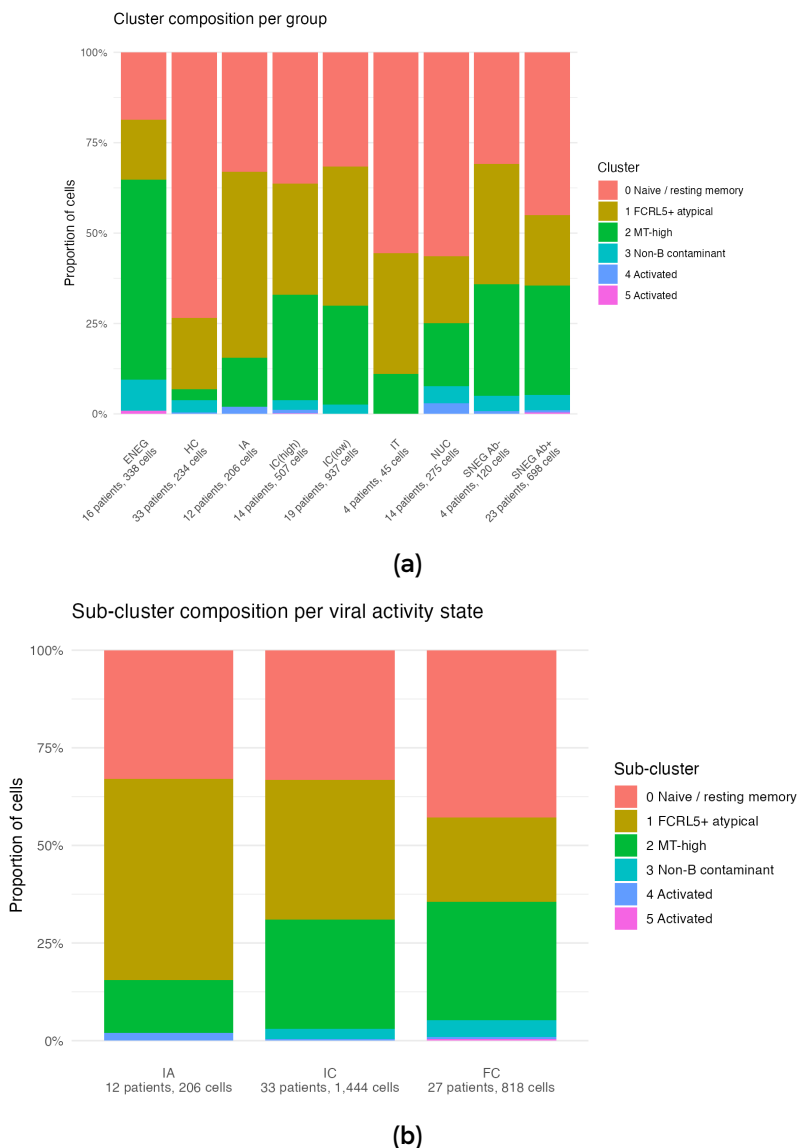


Figure 2.3. Subset composition (a) per clinical group and (b) per viral-activity state. Healthy controls are resting-dominated; the atypical fraction increases with viral activity (IA > IC > FC).

2.3 Differential expression: interferon with viral activity, ribosomal/secretory at cure, MHC-II versus healthy

In IA versus FC (Figure 2.4), the genes higher in immune-active cells form a coherent interferon-stimulated set (*IFI44L*, *XAF1*, *OAS1*, *EPSTI1*) together with the inhibitory check-point *VSIR* and the negative BCR-signaling adaptor *SLA*, while the genes higher in functional cure are ribosomal and translational (*RPS26*, *RPS27A*, *EEF1A1*) and antibody-secretion machinery (*JCHAIN*, *POU2AF1*, *IGLC2*). In IC versus FC the unswitched isotypes *IGHM* and *IGHD* and the B-cell-receptor kinase *SYK* are higher in IC. The largest contrast is IA versus HC (694 genes): immune-active cells are atypical (*ITGAX*, *SIGLEC6*, *FGR*), express a full MHC class II module (*HLA-DRA*, *HLA-DRB1*, *HLA-DPB1*, *CD74*, *IFI30*), and are interferon-high and class-switched, whereas healthy cells are resting (*SELL*, *FOXP1*) and ribosomal. Within the atypical cluster specifically, the interferon set (*IFI44L*, *XAF1*, *MX1*, *PARP9*, *SAMD9L*) is the cleanest signal higher in IA, localizing the interferon program to atypical cells. All of this is [cell-level] (Table 2.2).

Table 2.2. Significant differentially expressed gene counts (unadjusted $p < 0.05$, $|\log_2 FC| > 0.25$). The atypical cluster (1) carries the most genes on the activity axis; IA versus HC is the largest contrast. These are cell-level counts (see Table 2.3).

Contrast	whole	cluster 0	cluster 1	cluster 4
IA vs FC	224	265	444	51
IC vs FC	345	289	323	26
IA vs IC	255	312	243	58
FC vs HC	406	325	447	–
IA vs HC	694	567	420	–

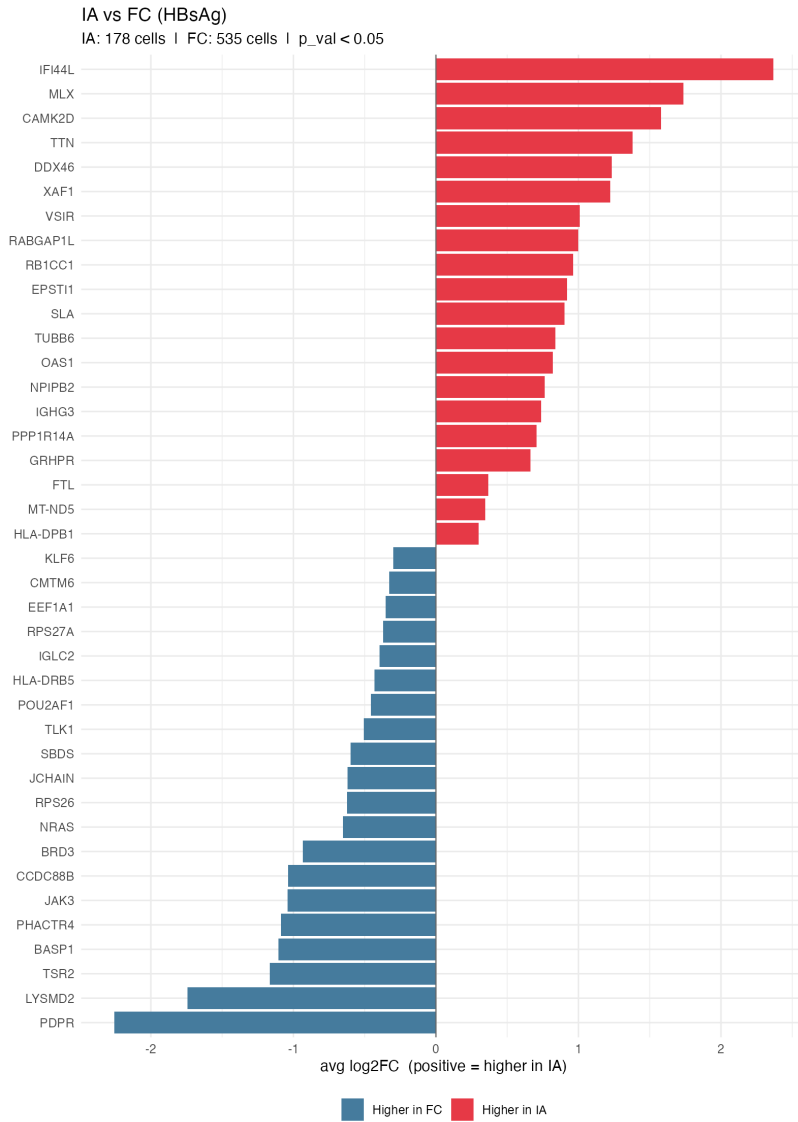


Figure 2.4. Top differentially expressed genes, IA versus FC. Interferon genes (*IFI44L*, *XAF1*, *OAS1*, *EPSTI1*) are higher in IA; ribosomal (*RPS26*, *EEF1A1*) and secretory (*JCHAIN*, *POU2AF1*) genes are higher in FC. Ribosomal genes appear here but are removed from the enrichment ranking, so this signal is visible only in the differential-expression figures.

2.4 Pathway enrichment recapitulates the themes but is dominated by pseudoreplication

The clearest enrichment is IA versus HC, where the top sets are all MHC class II antigen processing and presentation (Figure 2.5); IC versus FC is dominated by B-cell-receptor and B-cell-identity sets higher in IC. The single-sample enrichment differences are subtle with large distributional overlap, and crucially they do not survive patient-level testing: for IA versus FC, 388 Reactome pathways are significant per cell but zero per patient, and a patient-label permutation null shows the cell-level test is grossly anticonservative (Figure 2.6). The enrichment results are therefore [cell-level] hypotheses only.

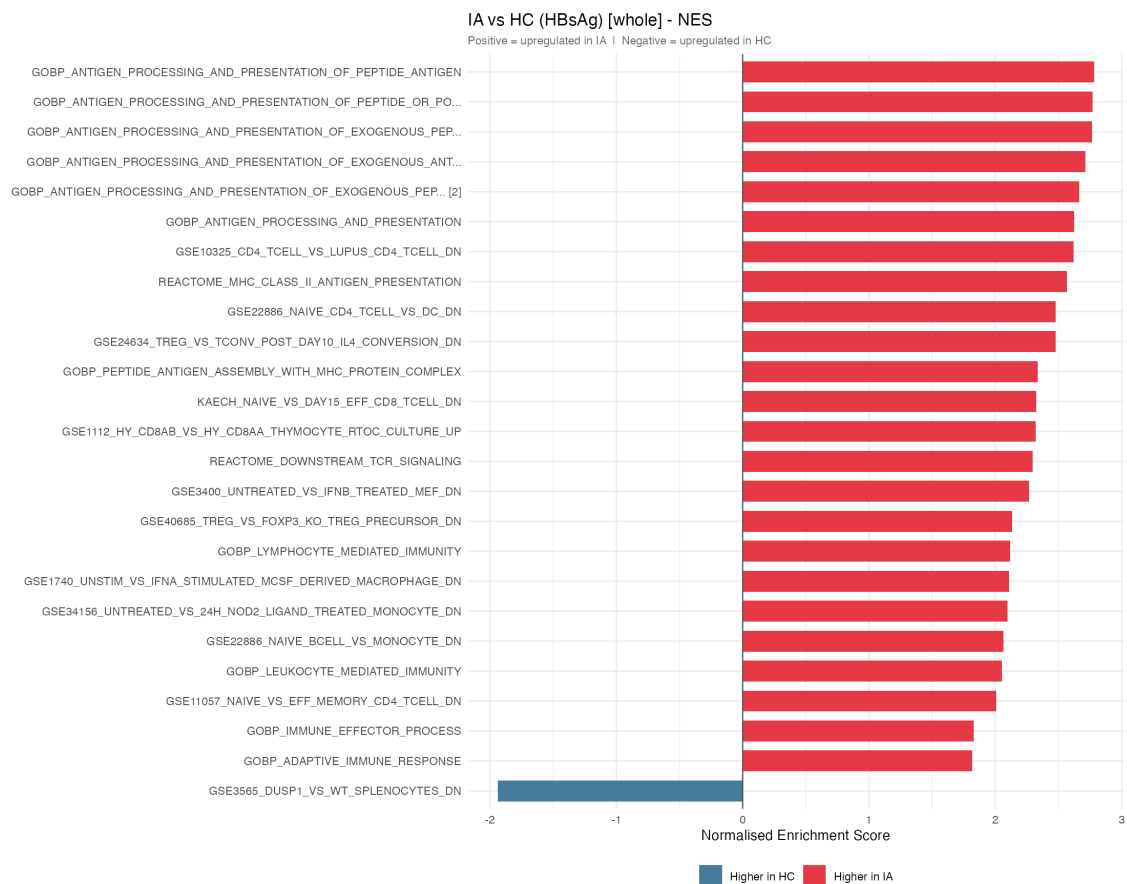


Figure 2.5. Gene-set enrichment, IA versus HC. MHC class II antigen processing and presentation dominates the sets enriched in immune-active cells relative to healthy; essentially everything is enriched in IA.

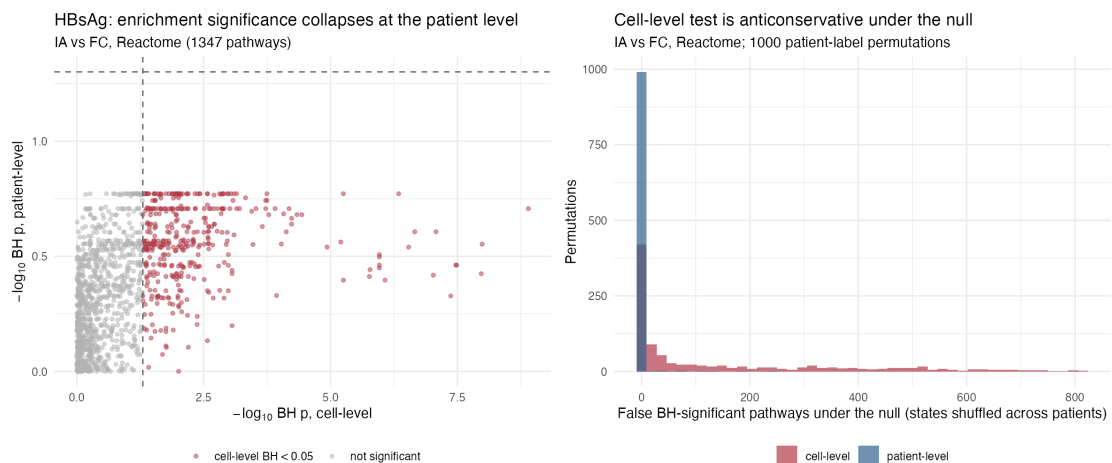


Figure 2.6. Pseudoreplication check (IA versus FC, Reactome). Left: every pathway significant at the cell level (red) falls below the patient-level significance line. Right: under a patient-label permutation null the cell-level test yields up to hundreds of false-positive pathways while the patient-level test stays at zero (Squair et al. 2021).

2.5 Only the interferon module survives the patient-level reality check

Six curated programs were scored per cell and re-tested with the patient as the unit (Table 2.3, Figure 2.7). The interferon (ISG) module is the one program that survives: higher in IA than FC (patient $p = 0.007$) and than HC (patient $p = 0.008$), both passing correction across the six modules, corroborated by the interferon genes above and localized to the atypical subset. **[patient-robust]** The atypical module (higher in IC than FC, and in IA than HC) and reduced T-follicular-help receptivity (in IC) reach only nominal patient-level significance (**[cell-level]**, suggestive). The resting, activation and TNF-NF- κ B modules do not differ robustly; their cell-level differences collapse entirely at the patient level. Functional-cure cells are essentially identical to healthy controls across all six modules, in sharp contrast to the strongly remodeled IA versus HC. **[descriptive]**

Table 2.3. Curated module scores: cell-level versus patient-level test. Only the interferon (ISG) module survives patient-level aggregation with correction, in both IA contrasts. Everything else is nominal or collapses.

Contrast	Module	Δ	cell p	patient p	Verdict
IA vs FC	isg	+0.114	6.9e-08	0.0067	survives
IA vs FC	atypical	+0.223	2.4e-07	0.081	nominal-miss
IC vs FC	atypical	+0.150	2.6e-06	0.026	nominal
IC vs FC	t-help receptivity	-0.041	7.4e-06	0.020	nominal
IA vs HC	isg	+0.086	1.9e-04	0.0079	survives
IA vs HC	atypical	+0.267	3.3e-09	0.062	nominal-miss
IA vs HC	resting	-0.181	1.3e-04	0.849	collapses
FC vs HC	(all six)	small	≥ 0.068	≥ 0.35	none

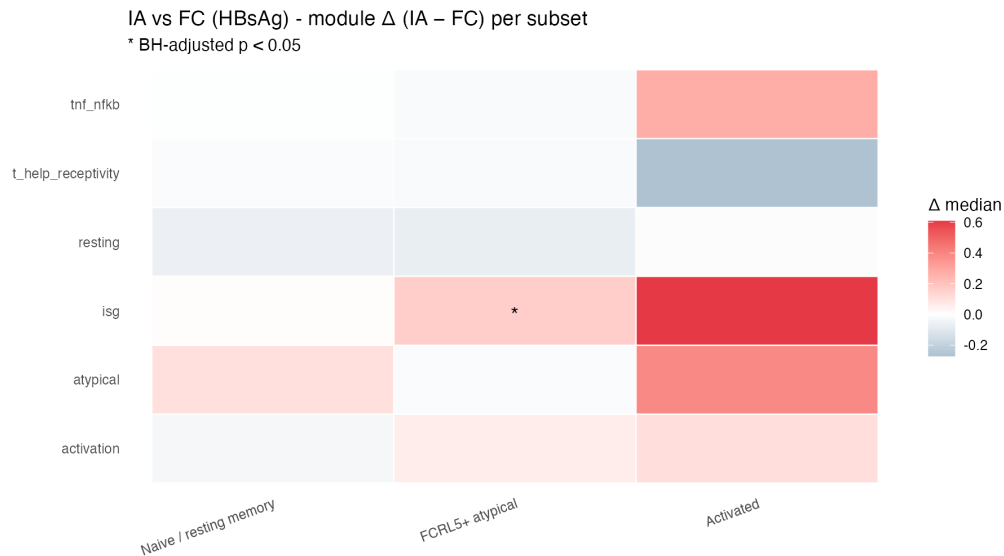


Figure 2.7. Module change (IA minus FC) by subset. The interferon (ISG) module is significantly higher in IA within the FCRL5⁺ atypical subset (starred); the interferon signal is carried by the atypical compartment.

2.6 Confounders and artifacts

Several issues limit interpretation. Pseudoreplication is the dominant one: most cell-level significance is an artifact of treating hundreds of cells from 12 to 33 patients as independent. A batch-effect check (output/diagnostics/batch_hbsag/) shows batch drives the excluded clusters (MT-high = batch 3) but not the analyzed resting/atypical clusters (Cramer's V 0.13), so no integration is needed. The whole-object contrasts were adjusted for sequencing batch (2 to 4 shared batches), but sequencing batch is moderately confounded with state (Cramer's V 0.35; IA cells fall only in batches 4 and 5), so batch and viral activity cannot be cleanly separated, and integration would risk removing real state signal. A few tiny per-cluster splits (the IA-versus-FC atypical cluster 1; the cluster-4 splits) shared only one batch and fell back to the unadjusted test, so that per-cluster interferon localization is not batch-adjusted; the excluded MT-high cluster also overlaps a single batch. Y-chromosome genes (*KDM5D*, *RPS4Y1*) among the IA-higher genes indicate a sex-composition imbalance between groups. Several recurrent hits are artifact-suspect and need a doublet/ambient check: *PCDH9* (recurrent, no B-cell anchor), *CAMK2D* (direction-inconsistent between contrasts), *PDPR* (recurrent on the lower-activity side), and the myeloid *CD68/SLC11A1*. [\[dataset-only, not literature-supported\]](#)

3

Discussion

3.1 One finding survives scrutiny: interferon rising with viral activity

The type-I interferon-stimulated-gene signature is the strongest result: it is the only module surviving patient-level aggregation, is built on canonical interferon genes, and localizes to the FCRL5⁺ atypical subset. Its direction has independent support from analogues, sustained type-I interferon generating an antigen-specific interferon-stimulated memory B cell subset in mouse chronic infection (Cooper et al. 2024), human chronic-HBV atypical B cells expressing IFNLR1 and interferon-stimulated genes (Osmani et al. 2024a), and interferon signatures peaking in the active phase (Vanwolleghem et al. 2021). Two caveats are essential, however. No human antigen-specific-sorted study has shown an activity-graded interferon program, so the antigen-specific, graded nature is an inference. And the same laboratory's total-B-cell study reports the opposite, no type-I interferon program in HBV B cells, with immune-active activation being an interferon-gamma and humoral program instead (Osmani et al. 2024b); a cross-disease study likewise attributes atypical differentiation to interferon-gamma (Holla et al. 2021), and the lab's quantitative-HBsAg work found type-I interferon genes in the HBsAg-low compartment, the opposite ordering in a different cell type (Beudeker et al. 2024). The parsimonious reconciliation is that total B cells are not antigen-specific B cells, and a subset feature is not a bulk feature: the signal is carried by a minority atypical population that bulk profiling dilutes. This divergence is presented as a hypothesis to test, not resolved in the dataset's favor.

3.2 Atypical dominance is supported; whether it tracks activity is contested

That the HBsAg-specific compartment is atypical-dominated and lacks a large activated cluster is well supported for this antigen: HBsAg-specific B cells are predominantly atypi-

cal and secretion-defective (Salimzadeh et al. 2018; Burton et al. 2018), more atypical than HBcAg-specific cells (Le Bert et al. 2020; Vanwolleghem et al. 2021), and the FCRL5/ITGAX/TBX21/ZEB2 core is the conserved atypical signature (Gao and Cockburn 2022; Yang et al. 2022; Holla et al. 2021); the high-ribosomal resting cluster is likewise a genuine resting-memory signature (Graus et al. 2025). The compositional gradient is directionally consistent with atypical cells expanding with antigen and contracting at HBsAg loss (Gu et al. 2024). But whether atypical frequency genuinely tracks the axis is disputed: it did not associate with clinical phase in the lab's own study (Osmani et al. 2024a), did not differ between HBsAg-high and -low patients (Kim et al. 2020), and single-cell profiling places the atypical subset at the tolerant rather than active end (Li et al. 2025); the atypical state is also near-irreversible, a poor phase tracker (Cooper et al. 2024; Poonia et al. 2018). With our own tracking statistic only nominal, the atypical-tracks-activity claim is best read as a contested trend.

3.3 Functional cure approaches, but may not fully equal, health

That functional-cure cells were near-identical to healthy controls on all six modules, while immune-active cells were strongly remodeled, matches the literature's partial-rescue framing: at HBsAg loss, antigen-specific B cells shift back toward resting memory yet retain residual dysfunction and are not fully normalized (Gu et al. 2024; Poonia et al. 2018). Our one residual difference, slightly elevated MHC class II (*HLA-DQB2*) and immunoglobulin genes in functional cure, is compatible with a partial reset, although *HLA-DQB2* as a functional-cure marker is [\[dataset-only, not literature-supported\]](#). The cure-approaches-health reading is supported in spirit but should be stated as partial resemblance.

3.4 The antigen-presentation signal is real in the data but its direction is contested

Immune-active cells showed a strong MHC class II program versus healthy. Increased presentation capacity in activated/atypical B cells is plausible, but the direction is contested: HBV was reported to downregulate antigen-presentation genes in patient B cells while upregulating inhibitory receptors (Poonia et al. 2018). Because the MHC-II result also flips direction between contrasts and does not robustly survive patient-level testing, it is best treated as a context-dependent, cell-level observation.

3.5 Observations without direct literature support

Three Results are honest dataset observations without a matching antigen-specific-B-cell source: the reduced T-follicular-help-receptivity module in IC (the related literature is T-

cell-side); the IC-versus-FC contrast of unswitched isotypes and B-cell-receptor signaling against ribosomal genes; and the secretory/plasma lean of functional-cure cells. Each is [dataset-only, not literature-supported].

3.6 Methodological limitations are the best-supported part of the report

The one finding with unambiguous literature backing is a caution: treating cells as replicates inflates single-cell differential-expression false positives, and aggregating to the patient is the fix (Squair et al. 2021; Crowell et al. 2020; Zimmerman et al. 2021). Our permutation null reproduces exactly this, which is why only the interferon module is carried forward as robust. Batch confounding (whole-object DEGs are batch-adjusted, but batch stays correlated with state and a few tiny per-cluster splits could not be adjusted), sex confounding, and the artifact-suspect genes above compound it, and are [dataset-only, not literature-supported] design limitations.

3.7 The recurring theme, and what a human should do next

Where this dataset diverges from prior work, the divergences cluster on two axes that are themselves the findings: total B cells are not antigen-specific B cells (Osmani et al. 2024b,a; Kim et al. 2020), and HBsAg-specific is not HBcAg-specific (Le Bert et al. 2020; Vanwolleghem et al. 2021) (the formal antigen comparison is left to future work). Because this is an autonomous demonstration, its outputs are hypotheses. Before any enters a scientific claim, a researcher should re-test the activity axis with pseudobulk or mixed-model statistics on the patient (Squair et al. 2021), balance or model sex and sequencing batch, validate the interferon signal and the atypical gradient at the protein level and in an independent cohort, and confirm that *PCDH9*, *CAMK2D* and the myeloid genes are not doublet artifacts. The single claim that would survive today is narrow: *in this dataset, a type-I interferon program is elevated in immune-active HBsAg-specific B cells relative to functional cure and healthy controls, concentrated in the atypical subset, robust to patient-level aggregation, and directionally consistent with, but not proven by, mouse and total/atypical human analogues.*

References

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